

Wine Quality Prediction - Project Report

Dataset

The project uses the Red Wine Quality Dataset containing 1,599 samples, 11 physicochemical features and one target (quality). The target is converted into Good (≥ 7) and Bad (< 7).

Features

Fixed Acidity, Volatile Acidity, Citric Acid, Residual Sugar, Chlorides, Free Sulfur Dioxide, Total Sulfur Dioxide, Density, pH, Sulphates and Alcohol.

Algorithms Used

Classification: Random Forest, SVM, KNN, Logistic Regression, Gradient Boosting. Clustering: K-Means, Agglomerative, DBSCAN.

EDA Observations

Most wines have quality 5-6. Higher alcohol is positively associated with quality while volatile acidity is negatively associated. Most correlations are weak to moderate.

Classification Results

Random Forest achieved the best overall performance (~94.1% accuracy, $F1 \approx 0.732$), followed by Gradient Boosting.

Clustering Results

K-Means produced the best overall clustering quality while DBSCAN performed poorly.

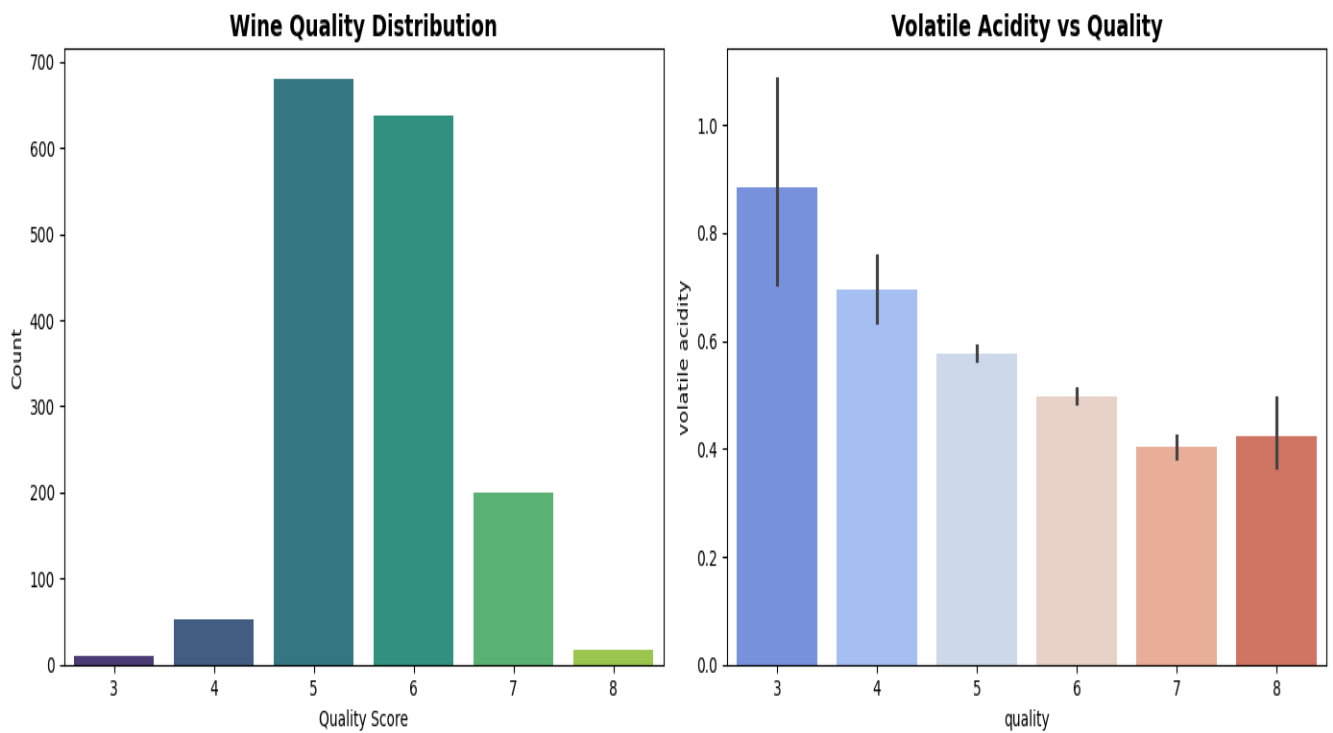
Conclusion

Random Forest is the best classifier and K-Means is the best clustering algorithm for this dataset.

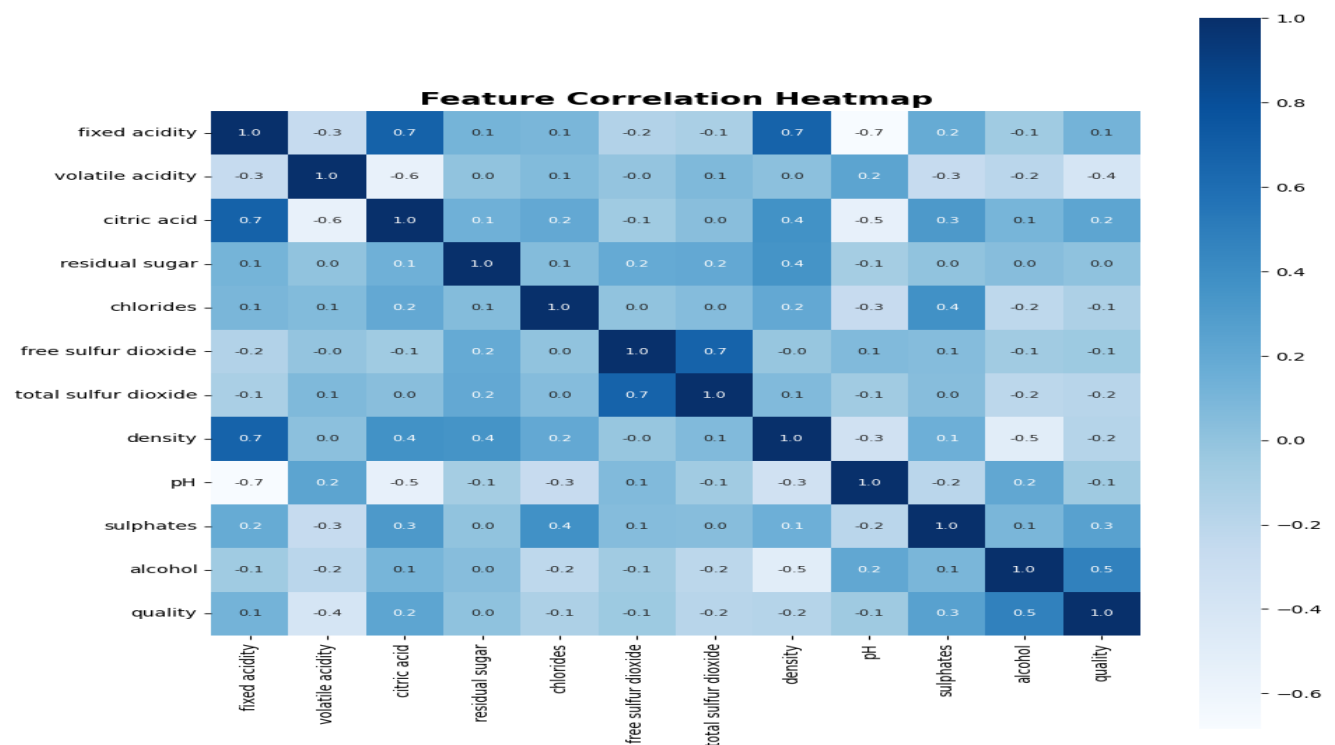
Future Improvements

Hyperparameter tuning, XGBoost/LightGBM/CatBoost, SHAP explainability and deployment.

Wine Quality Distribution & Volatile Acidity

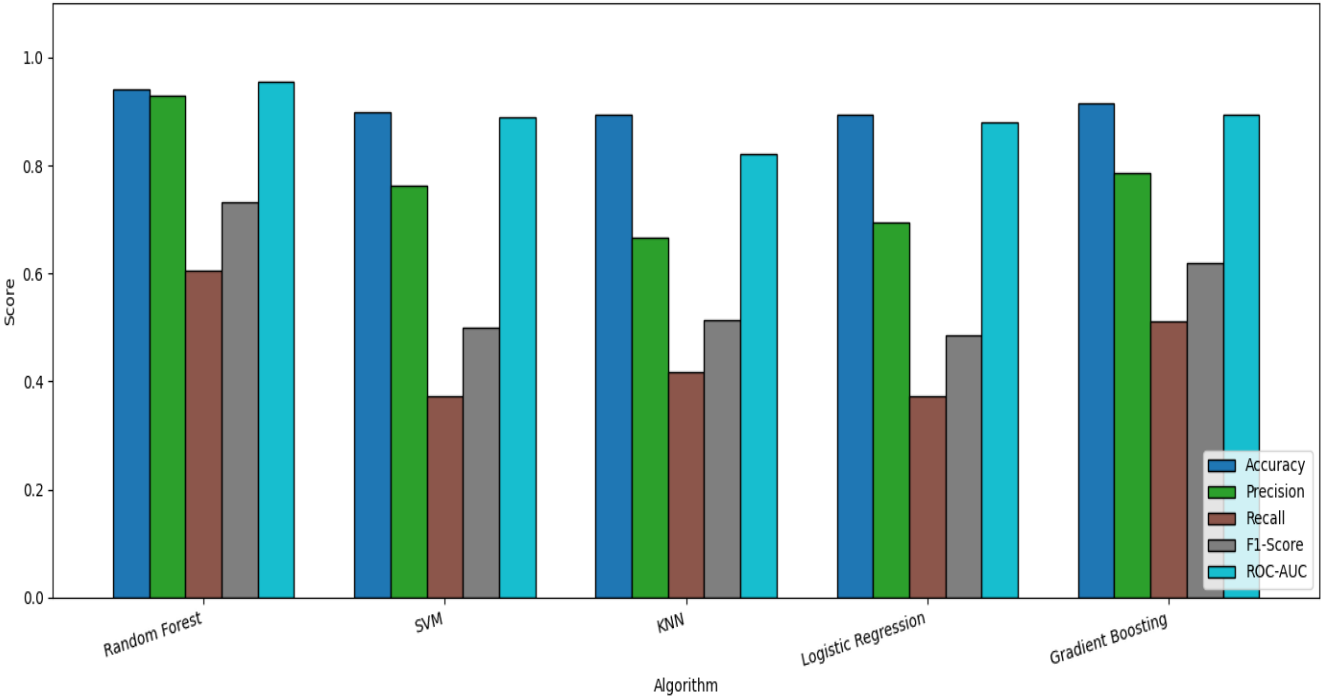


Feature Correlation Heatmap



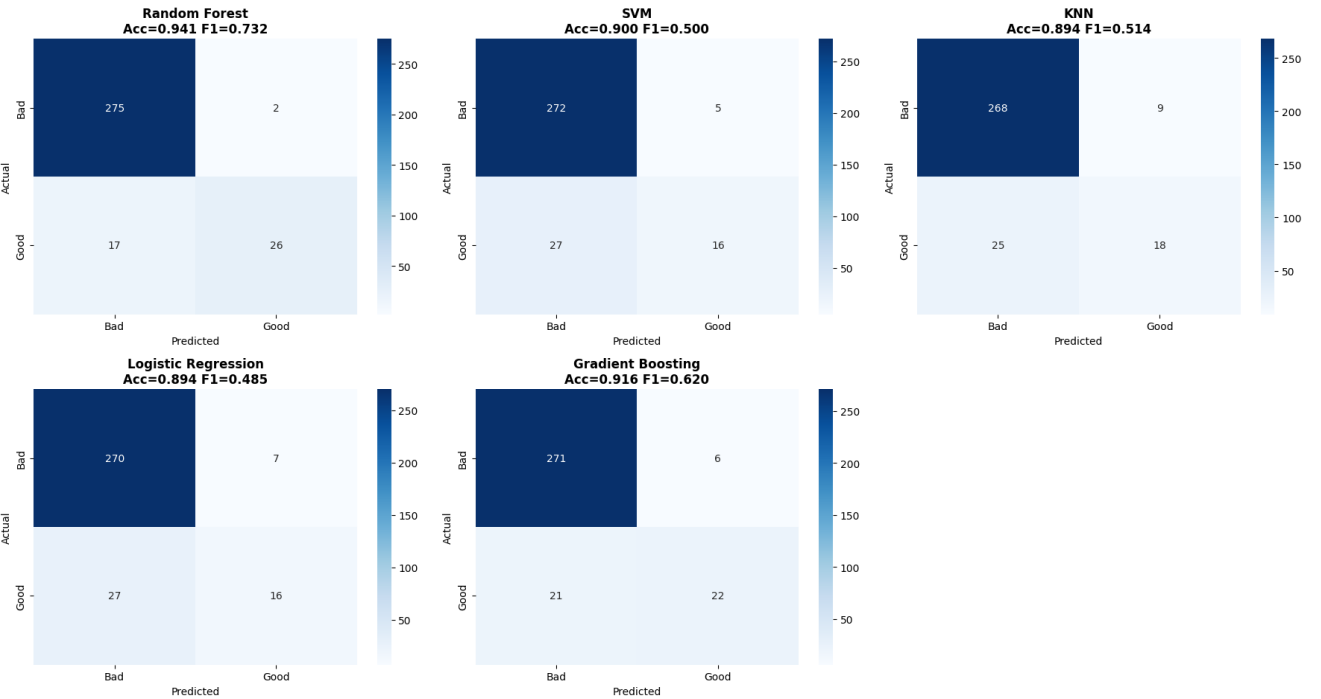
Classification Metrics Comparison

Classification Algorithms Metrics Comparison



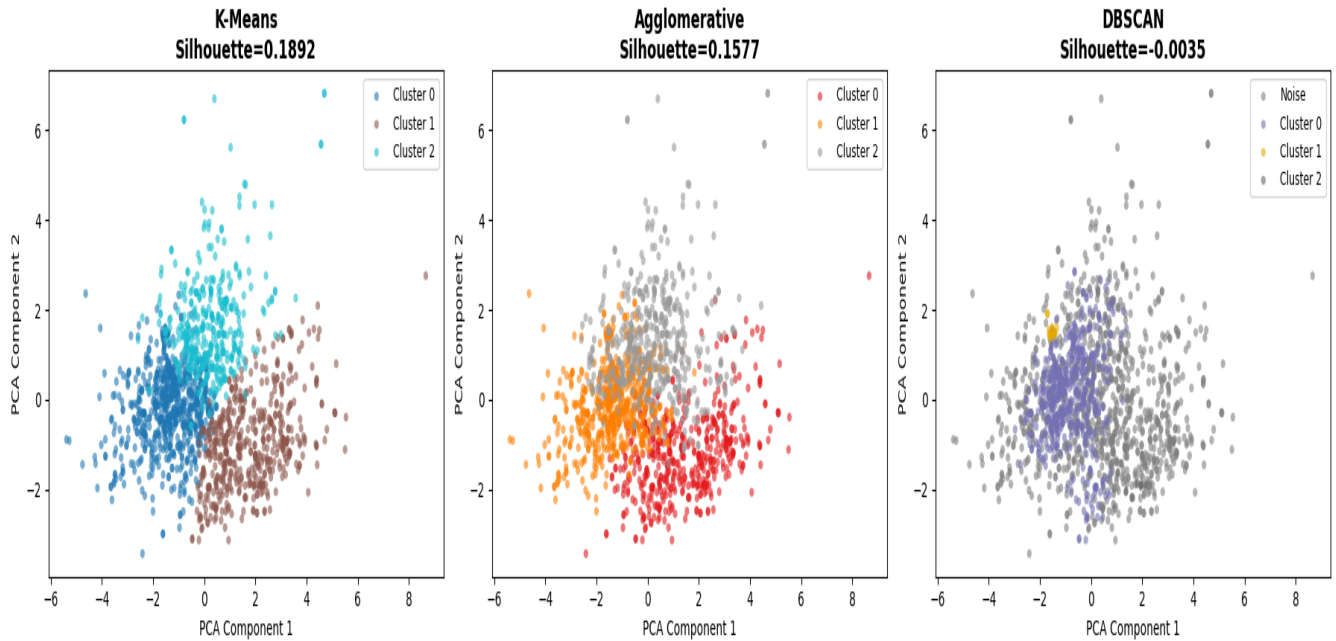
Confusion Matrices

Confusion Matrices — All Classifiers



Cluster Visualization

Cluster Visualization (PCA – 2D Projection)



Clustering Metrics Comparison

Clustering Evaluation Metrics Comparison

